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- 2
- 3
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- 8
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Abstract

Objective: To determine whether county-level social vulnerability is associated with the proportion of diagnosed diabetic retinopathy (DR) that has progressed to a vision-threatening stage, independent of county demographic composition, eye care supply, and spatial structure.

Design: Cross-sectional analysis of county-level surveillance estimates.

Subjects: Adults with diabetes residing in 3,119 United States counties and county-equivalents, representing 36.1 million adults with diabetes, 9.53 million with DR, and 1.82 million with vision-threatening DR (VTDR).

Methods: County estimates of DR and VTDR prevalence among adults with diabetes were obtained from the Centers for Disease Control and Prevention Vision and Eye Health Surveillance System (VEHSS) for 2021. Because VEHSS derives county estimates by post-stratification of national demographic-specific rates onto local population composition, all models were estimated *within* county $\times$ race/ethnicity $\times$ age strata with stratum fixed effects, so that county racial and age composition could not mechanically contribute to the estimated association. Social vulnerability was measured with the 2020 CDC/ATSDR Social Vulnerability Index (SVI). Geographic access was measured as road-network drive time from population-weighted county centroids to the nearest accredited ophthalmology residency program and to the nearest county containing any patient-care ophthalmologist. Models were quasibinomial with VTDR cases as successes and all-stage DR cases as trials. Residual spatial dependence was addressed with a Leroux conditional autoregressive model.

Main Outcome Measures: The VTDR share, defined as vision-threatening DR cases divided by all-stage DR cases among adults with diabetes.

Results: Nationally, 19.1% of diagnosed DR was vision-threatening. Within demographic strata, each standard deviation increase in county social vulnerability was associated with higher odds of vision-threatening disease among people with DR (odds ratio 1.036, 95% CI 1.023 to 1.049; $p < 0.001$, cluster-robust), corresponding to approximately 0.43 percentage points per standard deviation. The gradient was consistent across all six race-by-age strata examined (β 0.032 to 0.046). Among adults aged 65 to 84, who are near-universally Medicare eligible, the gradient was detectably but only modestly shallower than among adults aged 40 to 64, an attenuation of about

12% (interaction -0.0036 , $p = 0.002$). The association persisted under a Leroux conditional autoregressive model with strong residual spatial dependence ($\rho = 0.83$): SVI 0.021 (95% credible interval 0.012 to 0.029). The gradient was unchanged by adjustment for Medicare imaging, evaluation and management, and diagnostic test volume per 1,000 beneficiaries (0.035 to 0.035), arguing against differential detection, but was attenuated by approximately 42% by county mammography screening completion (0.035 to 0.021), implicating general preventive-screening engagement. County uninsured rate and dual-eligible share did not explain it. Estimated without stratification, the same association was 0.146, indicating that approximately 76% of the apparent county-level social gradient in published VEHSS estimates is attributable to demographic post-stratification rather than to local variation. Decomposition shows this is almost entirely *racial* composition: holding age fixed leaves the coefficient unchanged (0.150), whereas holding race fixed collapses it (0.034).

Conclusions: Among adults who already have diabetic retinopathy, those living in more socially vulnerable counties are more likely to have disease at a vision-threatening stage. The gradient is not explained by demographic composition, is only marginally narrowed by Medicare eligibility and is not explained by direct measures of insurance coverage or diagnostic intensity, and is distinct from the well-described association between social disadvantage and DR occurrence. A substantial part of it is shared with general preventive-screening engagement, suggesting that the operative failure is not ophthalmology-specific. Analyses of county-level VEHSS estimates that do not hold demographic composition fixed will substantially overstate local social gradients.

1 Introduction

Diabetic retinopathy (DR) remains a leading cause of vision impairment among adults in the United States.[2, 15] In 2021, an estimated 9.60 million people were living with DR, representing 26.43% of individuals with diabetes, and 1.84 million had vision-threatening DR (VTDR).[15]

The distinction between these two figures is clinically decisive and analytically underused. Whether a person with diabetes develops any retinopathy is largely a function of glycemic exposure, diabetes duration, and comorbid vascular risk. Whether that retinopathy is detected while still mild, or instead presents as severe non-proliferative disease, proliferative disease, or diabetic macular edema, is a function of whether the person was screened on schedule and retained in care.[2, 12, 21] The first is a question about disease occurrence; the second is a question about disease progression and timely detection.

A substantial literature establishes that adverse social determinants of health are associated with reduced eye care utilization and lower adherence to DR screening.[6, 16, 18] It is correspondingly well established that DR is more prevalent in socially disadvantaged populations. Because that association is largely mediated by the higher prevalence of diabetes itself in disadvantaged populations, it says comparatively little about the quality or timeliness of ophthalmic care those populations receive.

Recent work has begun to address disease severity directly. Xie and colleagues, studying 73,618 patients with diabetes at two academic centers, found that residents of more deprived neighborhoods had substantially higher odds of proliferative diabetic retinopathy, with a clear dose-response across quartiles of the Area Deprivation Index.[20] That study establishes the gradient in a large clinical population. Three questions it could not address motivate the present analysis. Its sample consists of patients attending tertiary referral centers, so its generalizability to the national population is unknown, as the authors note. Its outcome is proliferative disease among all patients with diabetes, which combines whether retinopathy developed at all with how far it progressed. And, as the authors also note, it could not adjust for neighborhood diabetes prevalence, leaving open whether the gradient reflects more diabetes or worse control of it.

The unresolved question is therefore whether social disadvantage is associated with how far disease has advanced *among people who already have it*, in the general population rather than among

referral-center attendees. That quantity, the share of diagnosed DR that is vision-threatening, is a case-mix measure rather than a burden measure. It is conditioned on disease, so it is largely orthogonal to diabetes prevalence, and it speaks directly to the adequacy of screening and follow-up rather than to the social patterning of diabetes.

Addressing this question with county-level surveillance data requires confronting a measurement problem that has not, to our knowledge, been addressed in prior applications of these data. The VEHSS county estimates are constructed by applying national age-, sex-, and race-specific prevalence rates to each county’s demographic composition, and then adjusting by a county random effect estimated from Medicare and Medicaid claims.[15] A county’s racial and age composition therefore moves its published estimate by construction. Any analysis relating those estimates to county-level social or demographic indices, including composite indices such as the Social Vulnerability Index that contain a racial and ethnic minority status component, will recover part of the post-stratification machinery rather than a local epidemiologic signal.

We therefore estimate the association between county social vulnerability and DR progression entirely within county $\times$ race/ethnicity $\times$ age strata, so that demographic composition is held fixed by design. We further ask whether any such gradient is narrowed by near-universal insurance coverage at age 65, whether it is explained by the supply of eye care providers or by geographic proximity to ophthalmology training infrastructure, and whether it survives adjustment for spatial dependence.

Figure 1 states the proposed causal pathway and the two non-causal paths our design is built to block.

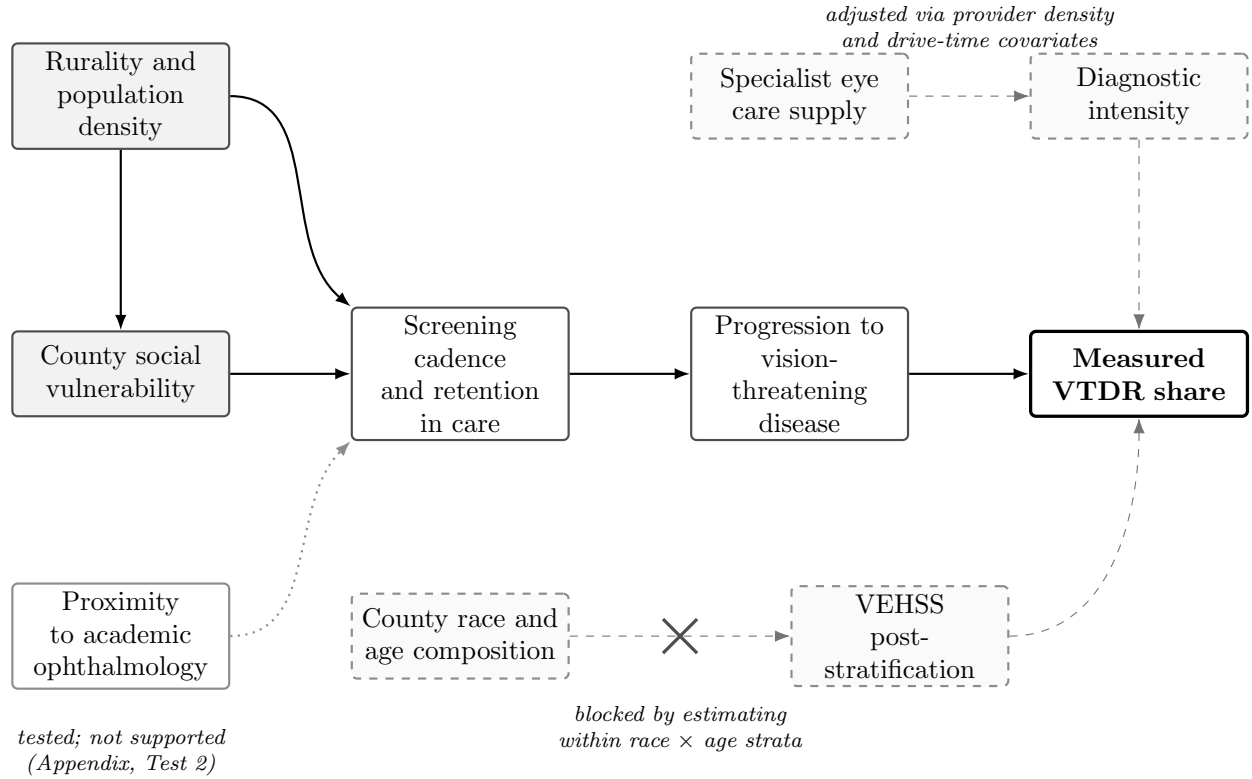


Figure 1: Proposed causal pathway from county social vulnerability to the measured vision-threatening share of diabetic retinopathy. Solid arrows denote the hypothesized causal chain. Dashed arrows denote non-causal paths that would induce an association in the absence of any true effect: demographic post-stratification in the construction of the outcome (blocked by design through within-stratum estimation) and differential detection driven by specialist supply (addressed by covariate adjustment). The dotted arrow denotes the proximity-to-academic-ophthalmology pathway posited in prior work, which we tested and did not find support for (Appendix, Test 2).

2 Methods

This study follows the STROBE reporting guideline for cross-sectional studies; a completed checklist is provided as a supplementary file.

2.1 Study design and setting

We conducted a cross-sectional analysis of county-level surveillance estimates for the United States in 2021. All data were publicly available, aggregated, and contained no individual-level identifiers, so institutional review board approval was not required.

2.2 Units of analysis and eligibility

The unit of analysis was the county or county-equivalent jurisdiction. Of 3,143 counties in the 50 states and the District of Columbia, 3,119 had complete data on all analytic variables and constituted the analytic sample. Puerto Rico is not represented in the VEHSS source data and was necessarily excluded. Connecticut is represented by its eight pre-2022 counties, consistent with the 2021 vintage of all constituent data sources.

2.3 Outcome

The primary outcome was the *VTDR share*: the number of people with vision-threatening DR divided by the number with DR of any stage, among adults with diabetes. Both quantities were obtained from the CDC Vision and Eye Health Surveillance System composite prevalence estimates for 2021.[4, 15] VEHSS defines vision-threatening DR as severe non-proliferative DR, proliferative DR, or diabetic macular edema, and non-vision-threatening DR as mild to moderate non-proliferative or unspecified DR without vision-threatening features.[15]

We used the published case counts and the corresponding diabetes-population denominators rather than precomputed rates. Aggregating our county-level numerators reproduced the published national estimates: DR 26.43% of adults with diabetes, VTDR 5.05%, and a VTDR share of 19.1%, against published values of 26.43%, 5.06%, and 19.2% respectively.

2.4 Handling of post-stratification in the outcome

VEHSS county estimates are derived by computing national prevalence by age, sex, and race, applying those rates to each county’s population distribution across the same categories, and adjusting by a county random effect estimated from 2018 Medicaid claims and 2017 to 2019 Medicare Part B fee-for-service claims.[15] County demographic composition therefore enters the published estimate mechanically.

To remove this pathway, all models were estimated within county $\times$ race/ethnicity $\times$ age cells. We obtained separate VEHSS estimates for non-Hispanic White, non-Hispanic Black, and Hispanic adults, each in the 40 to 64 and 65 to 84 year age bands, giving six strata and 13,035 analytic cells across 3,111 counties. Eight of the 3,119 counties in the analytic sample have populations too small for VEHSS to publish race-stratified estimates and therefore contribute no cells; they are retained in county-level descriptive analyses only.

Cell coverage is 13,035 of a possible 18,666. Missingness is driven by the size of the demographic group rather than by any property of the outcome: VEHSS does not publish an estimate where the county’s population in that race-by-age cell is very small. Among counties with no non-Hispanic Black estimate for ages 65 to 84, the median non-Hispanic Black population is 105, against 5,576 in counties where the estimate is published, and 95% have fewer than 1,000. The corresponding medians for Hispanic adults are 420 against 6,128. The Black and Hispanic strata are therefore estimated on counties with substantial minority populations, which skew urban, and stratum-specific estimates should be read with that in mind. Models include stratum fixed effects, so the social vulnerability coefficient is identified only from differences between counties within the same race-by-age cell. County demographic composition cannot contribute to it.

We report the unstratified estimate alongside the stratified estimate to quantify the magnitude of the artifact.

2.5 Exposure and covariates

Social vulnerability was measured using the overall percentile ranking (RPL_THEMES) of the 2020 CDC/ATSDR Social Vulnerability Index,[5] which is constructed from 2016–2020 American Community Survey data and uses the same Connecticut county geography as the rest of our data.

Geographic access was characterized by two measures. First, drive time from each county’s population-weighted centroid to the nearest accredited ophthalmology residency program. Population-weighted centroids were taken from the Census Bureau’s 2020 Centers of Population file,[19] which locates the centroid where people actually live rather than at the geometric center of the county polygon. Program locations were compiled from publicly available FRIEDA/ACGME listings, geocoded, and assigned to counties by point-in-polygon overlay against 2021 TIGER/Line boundaries; 123 programs accredited in or before 2021, located in 90 counties, were retained. Drive times were computed over the road network using the Open Source Routing Machine. Second, and to distinguish academic proximity from eye care availability in general, we computed drive time to the nearest county containing at least one patient-care ophthalmologist.

Eye care workforce counts were obtained from the Area Health Resources File 2022–2023 release, which carries 2021-vintage counts,[11] and expressed as patient-care ophthalmologists and optometrists per 100,000 residents. County population, age structure, and racial and ethnic composition came from the 2021 American Community Survey 5-year estimates. Population density used TIGER/Line land area, addressing the concern that equally populous counties differing greatly in area do not carry equal burden density. Rurality was measured with the 2013 Rural-Urban Continuum Code.

2.6 External validation data

Three data systems independent of VEHS were used to probe the two principal alternative explanations, differential detection and insurance coverage.

From the CMS Medicare Geographic Variation Public Use File (county level, 2021)[9] we obtained, for the Original Medicare population, imaging events, evaluation and management events, and diagnostic test events per 1,000 beneficiaries, and the share of beneficiaries dually eligible for Medicaid. The three utilization measures index how much diagnostic activity actually occurs in a county, from claims rather than from provider counts, and are a substantially stronger control for differential detection than provider density. Dual eligibility identifies poverty *within* the fully insured 65 and over population.

From County Health Rankings 2021[8] we obtained the county uninsured rate, mammography screening completion, influenza vaccination, and preventable hospital stays (ambulatory-

care-sensitive admissions). Mammography completion is a Medicare-claims measure of whether a county's population follows through on a recommended preventive screen and has no ophthalmic content; it serves as an index of general preventive-care engagement. Preventable hospital stays measure ambulatory care failure in a different organ system and a different data system, providing a convergent-validity comparison.

Medicaid expansion status as of 2021 was taken from the Kaiser Family Foundation tabulation.[10] Twelve states had not adopted expansion; Missouri and Oklahoma implemented mid-year and were excluded from the expansion contrast rather than forced into either group. Because expansion altered coverage only for adults under 65, the 65 to 84 band provides a built-in placebo: any apparent modification of the social gradient by expansion status that appears equally in the 65 to 84 band cannot be a coverage effect.

2.7 Statistical analysis

Because the outcome is a count of vision-threatening cases out of a count of total DR cases, we fitted binomial-family generalized linear models with VTDR cases as successes and all-stage DR cases as trials. County aggregates of this kind are overdispersed, so all models use the quasibinomial family, which inflates standard errors by the estimated dispersion rather than treating each county as an independent Bernoulli sequence.

The primary model regressed the VTDR share on standardized SVI, drive time to the nearest residency program, drive time to the nearest ophthalmologist, ophthalmologist and optometrist density, population density, rurality, and stratum fixed effects.

Because the analytic file stacks up to six race-by-age cells per county, and every county-level covariate is repeated across a county's cells, the quasibinomial dispersion parameter alone is not an adequate treatment of the resulting dependence: it rescales all standard errors by a single scalar without modeling the within-county correlation, and the effective sample size for a county-level covariate is nearer the 3,111 counties than the 13,035 cells. All standard errors and confidence intervals for county-level covariates are therefore cluster-robust, clustered on county. This inflates them roughly fourfold relative to the model-based values and is the basis for every interval reported below. Continuous predictors were standardized, so coefficients are per standard deviation and directly comparable. Drive times were log-transformed.

221 To test whether the gradient reflects insurance coverage, we compared the SVI coefficient be-
222 tween the 40 to 64 and 65 to 84 age bands within race, using a two-sample z test on the difference.
223 Medicare eligibility at 65 makes coverage near-universal in the older band; if the gradient operates
224 through insurance, it should attenuate there.

225 Residual spatial autocorrelation was assessed with Moran’s I on a queen contiguity weights
226 matrix. Because dependence was substantial, we refitted the largest stratum with a Leroux con-
227 ditional autoregressive spatial random effect using CARBayes,[13] which accepts binomial counts
228 and trials directly. Three island counties had no queen neighbor and were linked to their nearest
229 county so that the adjacency graph was connected. The model was fitted with a burn-in of 20,000
230 and 140,000 iterations thinned by 20, yielding 6,000 retained samples; convergence was assessed
231 with Geweke diagnostics (all $|z| < 1.3$) and effective sample sizes (576 to 3,832).

232 Spearman correlations are reported throughout in preference to Pearson, given the skewness of
233 county health and workforce distributions.

234 Analyses were conducted in R version 4.6.0. All data acquisition and analysis code is publicly
235 available.

3 Results

3.1 Sample characteristics

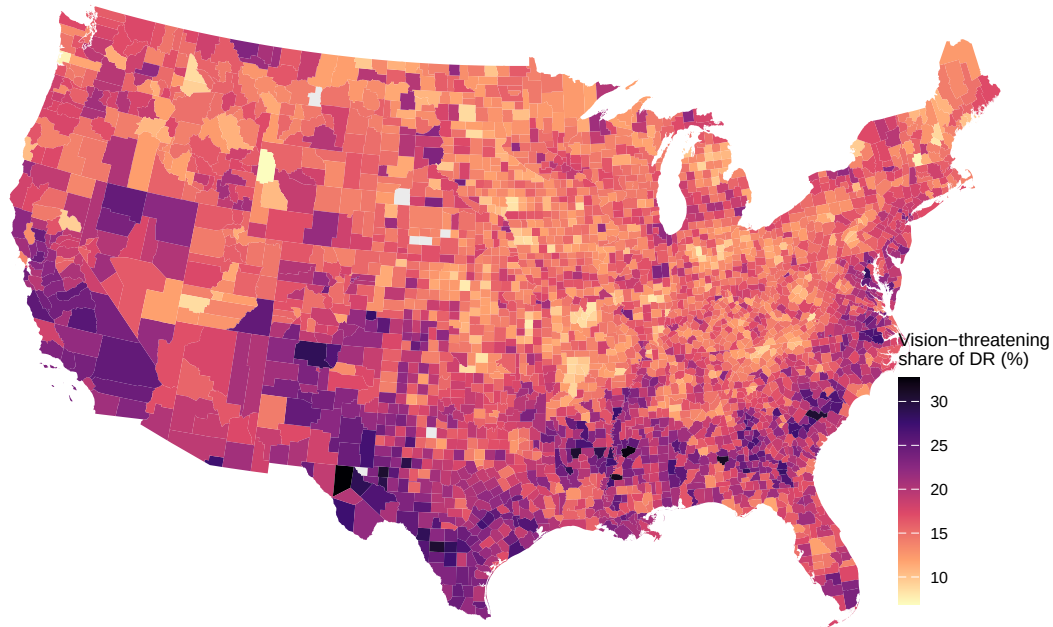
The analytic sample comprised 3,119 counties containing 36.1 million adults with diabetes, of whom 9.53 million had DR and 1.82 million had vision-threatening DR. Nationally, 19.1% of diagnosed DR was vision-threatening, but this share varied more than fourfold across counties, from 6.9% to 32.6% (interquartile range 14.0% to 19.2%).

The VTDR share was essentially uncorrelated with DR prevalence itself (Spearman $\rho = -0.08$), confirming that disease progression and disease occurrence are distinct county-level phenomena rather than two expressions of the same underlying gradient.

Median drive time to the nearest ophthalmology residency program was 123 minutes (interquartile range 77 to 192). Eighty-eight counties in the analytic sample contained a program. Notably, 1,950 of 3,119 counties (62.5%) contained no patient-care ophthalmologist of any kind.

Geography of diabetic retinopathy severity, not burden

Share of diagnosed DR that is vision-threatening, among adults with diabetes, 2021



Continental US shown; Alaska, Hawaii and Puerto Rico omitted. Darker counties have a higher proportion of DR that has reached a vision-threatening stage. This is a case-mix measure and is nearly uncorrelated with DR prevalence itself.

Figure 2: Geographic distribution of the vision-threatening share of diagnosed diabetic retinopathy. This is a case-mix measure and is nearly uncorrelated with diabetic retinopathy prevalence itself (Spearman $\rho = -0.08$).

3.2 Post-stratification accounts for most of the apparent social gradient

Estimated without holding demographic composition fixed, each standard deviation of county social vulnerability was associated with a coefficient of 0.146 on the log-odds of vision-threatening disease among people with DR. Estimated within race-by-age strata, the same association was 0.035. Approximately 76% of the apparent county-level social gradient in published VEHS estimates is therefore attributable to demographic post-stratification rather than to local variation (Table 1).

The artifact is specifically a *racial* composition effect, not an age effect. Holding age fixed while allowing racial composition to vary leaves the coefficient essentially unchanged at 0.150; holding race fixed while allowing age to vary collapses it to 0.034, which is already the fully stratified value. This is what the post-stratification mechanism predicts, since national race-specific vision-threatening rates differ substantially while the county-level social gradient is largely orthogonal to age composition.

This has direct implications for the interpretation of prior and future work using these widely used county estimates, and we return to it in the Discussion.

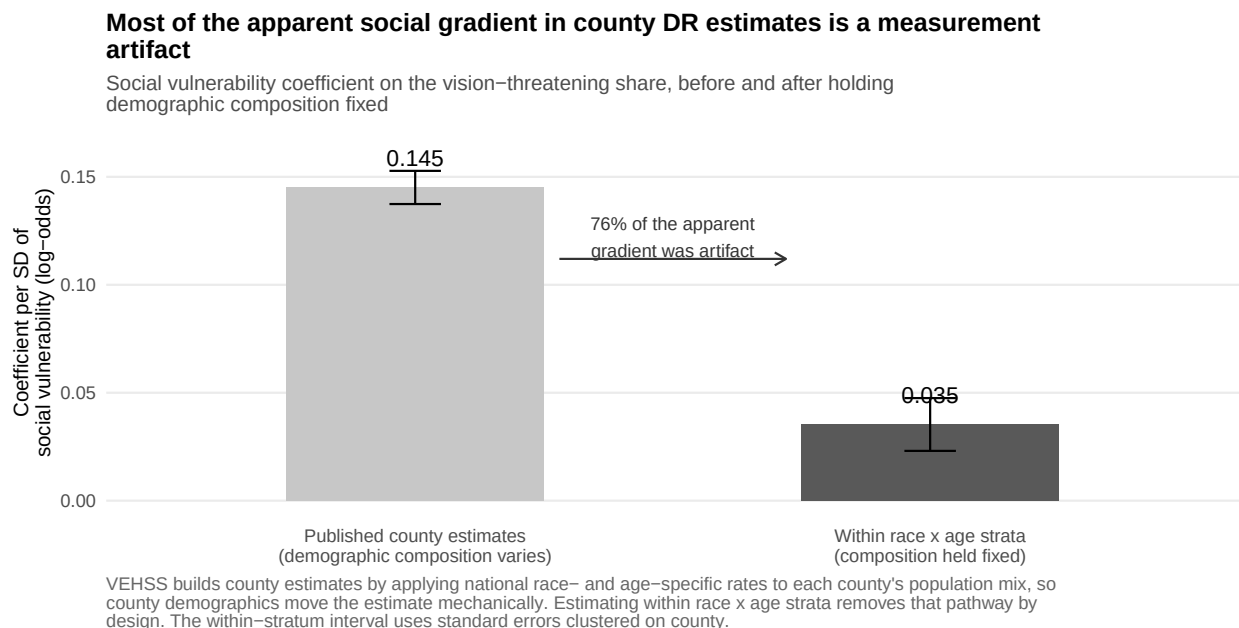


Figure 3: The social vulnerability coefficient before and after holding county demographic composition fixed.

3.3 Social vulnerability and disease progression

Within demographic strata, higher county social vulnerability remained associated with a greater vision-threatening share of diagnosed DR (Table 1). Each standard deviation increase in SVI corresponded to an odds ratio of 1.036 (95% CI 1.023 to 1.049; $p < 0.001$, with standard errors clustered on county), equivalent to approximately 0.43 percentage points of VTDR share per standard deviation at the non-Hispanic White 65 to 84 baseline of 14.2%.

The gradient was strikingly consistent across strata, ranging from 0.032 among non-Hispanic White adults aged 65 to 84 to 0.046 among Hispanic adults aged 40 to 64, and was statistically significant in every stratum (Table 2). That the same association appears at similar magnitude in three racial and ethnic groups and two age bands argues against its being an artifact of any single population.

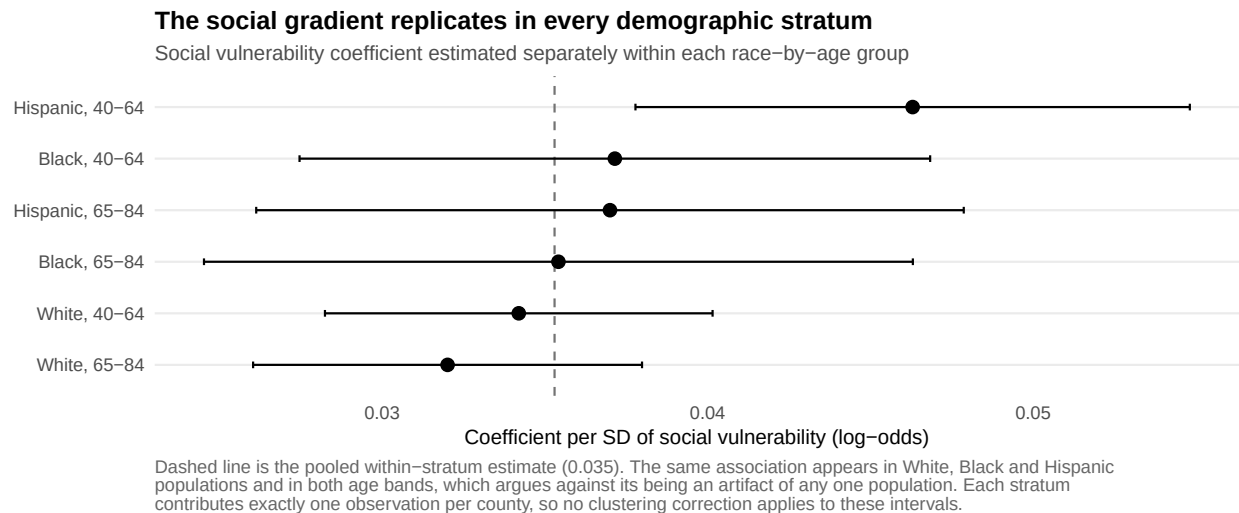


Figure 4: Social vulnerability coefficient estimated separately within each race-by-age stratum.

3.4 The gradient is not narrowed by Medicare eligibility

If the social gradient in disease progression operated principally through insurance coverage, it should be attenuated among adults aged 65 to 84, for whom Medicare coverage is near-universal. It was narrower, but only slightly. Estimated within demographic strata, the SVI coefficient was 0.037 among adults aged 40 to 64 and 0.033 among adults aged 65 to 84. Because the two age bands are drawn from the same counties, the correct test is an interaction within a single model

rather than a comparison of separately fitted coefficients; that interaction is -0.0036 ($p = 0.002$, cluster-robust), an attenuation of approximately 12%. Medicare eligibility is thus associated with a detectably but modestly shallower gradient, leaving roughly seven eighths of it intact. The same pattern held in each race stratum separately (Table 2).

3.5 Geographic access

Neither measure of geographic access behaved as an access hypothesis would predict. In the primary within-stratum model, greater drive time to the nearest residency program was associated with a *lower*, not higher, vision-threatening share ($\beta = -0.017$, $p < 0.001$), and drive time to the nearest ophthalmologist was not associated with the outcome ($\beta = -0.002$, $p = 0.52$). Detailed tests of the academic proximity hypothesis, including capacity-weighted access, discordant-county comparisons, and university versus community program contrasts, are reported in the Appendix; none supported it.

Two covariates that appeared significant under model-based standard errors do not survive clustering: ophthalmologist density ($p = 0.36$) and optometrist density ($p = 0.19$). Neither is a study exposure, but the change is worth noting, since it means eye-care provider density has no detectable independent association with disease severity once within-county dependence is accounted for. Rurality and population density remain associated (Table 1).

3.6 Sensitivity to the racial component of SVI and to provider density

Because the overall SVI contains a racial and ethnic minority status theme, we refitted the primary model using a composite of the remaining three themes, constructed as the unweighted arithmetic mean of the county percentile rankings for Theme 1 (socioeconomic status), Theme 2 (household characteristics), and Theme 4 (housing type and transportation), that is `RPL_THEME1`, `RPL_THEME2` and `RPL_THEME4`, then standardized to mean zero and unit variance in the same way as the overall index. It correlates with the overall SVI at 0.98. The association persisted at 0.027 (95% CI 0.015 to 0.040, $p < 0.001$), about three quarters of the overall-SVI estimate. Entered alone, the socioeconomic status theme gave 0.034 (95% CI 0.021 to 0.047).

The minority status theme was nonetheless the strongest single component, whether entered alone (0.069) or jointly with the other three (0.071). We emphasize that this is not the post-

stratification artifact, which within-stratum estimation removes by construction; it is a residual county-level association between racial composition and disease severity within demographic strata, and is consistent with the effects of residential segregation on care delivery rather than with a compositional artifact.

Because VEHSS already adjusted its county random effects for ophthalmologists per capita, our adjustment for ophthalmologist and optometrist density is partly a second pass at the same factor. Removing those two terms moved the SVI coefficient from 0.035 to 0.038, a change of 7%, so the double adjustment is not materially affecting the result.

3.7 Spatial confirmation

Residual spatial dependence was substantial (Moran's $I = 0.237$, $p < 0.001$). Under a Leroux conditional autoregressive model with strong estimated spatial dependence ($\rho = 0.82$), the social vulnerability association persisted with a posterior mean of 0.021 (95% credible interval 0.012 to 0.029). Drive time to the nearest residency program retained a small negative posterior mean whose credible interval marginally excluded zero (-0.010 , 95% CrI -0.019 to -0.001); that is, counties *farther* from a residency program had slightly *less* vision-threatening disease, the opposite of the direction an access hypothesis predicts. Optometrist density and rurality also retained credible intervals excluding zero, while drive time to the nearest ophthalmologist, ophthalmologist density, and population density did not (Table 3).

3.8 The gradient is unmoved by direct measures of diagnostic intensity

Adjusting for Medicare imaging, evaluation and management, and diagnostic test events per 1,000 beneficiaries left the social gradient essentially unchanged (0.0353 unadjusted to 0.0354 adjusted; Table 4). The detection pathway is nonetheless visible in these data: counties with more Medicare imaging had a *lower* vision-threatening share ($\beta = -0.028$, $p < 0.001$), consistent with greater diagnostic activity identifying more mild disease. That the social gradient is unaffected by adjustment for this pathway, while the pathway itself is measurable, is the strongest evidence available to us that the gradient is not a detection artifact.

3.9 General preventive-screening engagement accounts for a substantial share

Adjusting for county mammography screening completion attenuated the social gradient from 0.0353 to 0.0206, approximately 42% on the log-odds scale and 29% on the standardized scale (Table 4). Counties whose populations completed mammography screening more often had a lower vision-threatening share of diabetic retinopathy ($\beta = -0.039$, $p < 0.001$). Social vulnerability and mammography completion were correlated at -0.42 . Influenza vaccination added little beyond mammography.

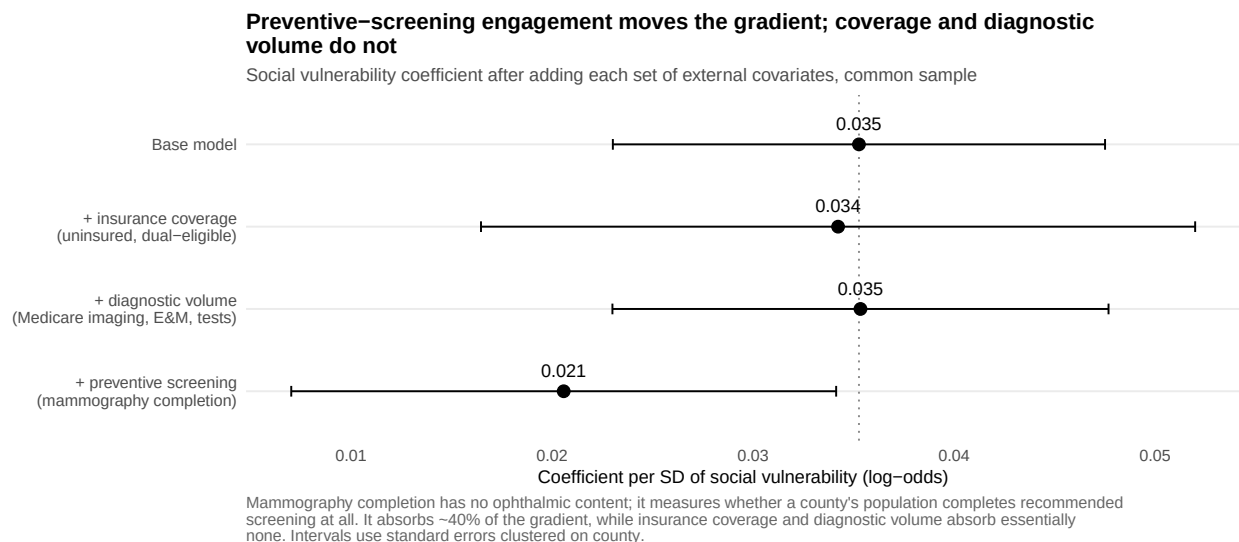


Figure 5: Social vulnerability coefficient after adding each set of external covariates on a common sample. Insurance coverage and diagnostic volume leave the gradient intact; preventive-screening completion absorbs roughly two fifths of it.

3.10 Direct measures of coverage do not explain the gradient

Adding the county uninsured rate and the dual-eligible share changed the gradient only slightly (0.0353 to 0.0343), and neither covariate was independently associated with the outcome (uninsured $p = 0.14$; dual-eligible $p = 0.44$). This confirms with direct measures what the age contrast showed indirectly. Within the 65 to 84 band alone, however, where Medicare coverage is near-universal, a higher dual-eligible share was still associated with a greater vision-threatening share ($\beta = 0.006$, $p < 0.001$), indicating that poverty continues to matter after coverage is essentially universal.

3.11 Medicaid expansion: an apparent effect that the placebo refutes

The social gradient was roughly three times steeper in expansion states than in non-expansion states among adults aged 40 to 64 (0.046 vs 0.016). Taken alone this would suggest expansion modifies the gradient. It does not survive the placebo. The same contrast appeared, essentially unchanged, among adults aged 65 to 84 (0.042 vs 0.012), an age band whose coverage expansion could not have altered, and the formal triple interaction was null ($\text{SVI} \times \text{expansion} \times \text{age band}$, $\beta = 0.002$, $p = 0.47$; Table 5). The apparent difference is therefore state-level confounding between expansion and non-expansion states rather than a coverage effect.

3.12 Convergent validity

Social vulnerability predicted ambulatory care failure measured entirely outside VEHSS: each standard deviation was associated with 0.279 standard deviations more preventable hospital stays ($p < 0.001$) and 0.465 standard deviations lower mammography completion ($p < 0.001$). The corresponding standardized association with the vision-threatening share was 0.100 ($p < 0.001$). The diabetic retinopathy progression gradient is thus real but modest relative to other county-level care-quality gradients, and sits within a coherent pattern of social gradients in ambulatory care.

Table 1: Within-stratum model of the vision-threatening share of diabetic retinopathy, 13,035 county $\times$ race $\times$ age cells across 3,111 counties. Quasibinomial generalized linear model with stratum fixed effects. Coefficients are per standard deviation on the log-odds scale.

Variable	β	Cluster SE	p	95% CI
Social Vulnerability Index	0.0353	0.0062	< 0.001	0.023 to 0.048
Drive time to residency program (log)	-0.0174	0.0053	0.001	-0.028 to -0.007
Drive time to any ophthalmologist (log)	-0.0018	0.0071	0.80	-0.016 to 0.012
Ophthalmologists per 100,000	-0.0072	0.0078	0.36	-0.022 to 0.008
Optometrists per 100,000	-0.0113	0.0085	0.19	-0.028 to 0.005
Population density (log)	-0.0405	0.0117	< 0.001	-0.063 to -0.018
Rural-Urban Continuum Code	-0.0584	0.0107	< 0.001	-0.079 to -0.037

Stratum fixed effects included. Dispersion parameter 1.93. Standard errors are cluster-robust, clustered on county (3,111 clusters); model-based SEs would be roughly four times smaller.

Unstratified comparison estimate for SVI: 0.146.

Table 2: Social vulnerability gradient in the vision-threatening share, estimated separately within each race-by-age stratum.

Stratum	n counties	β_{SVI}	SE	p
Non-Hispanic White, 40–64	3,096	0.0342	0.0030	< 0.001
Non-Hispanic White, 65–84	3,106	0.0320	0.0030	< 0.001
Non-Hispanic Black, 40–64	1,799	0.0372	0.0049	< 0.001
Non-Hispanic Black, 65–84	1,533	0.0354	0.0056	< 0.001
Hispanic, 40–64	2,060	0.0463	0.0043	< 0.001
Hispanic, 65–84	1,441	0.0370	0.0055	< 0.001

Table 3: Leroux conditional autoregressive model, non-Hispanic White adults aged 65 to 84, 3,106 counties. Posterior means and 95% credible intervals.

Variable	Posterior mean	95% CrI
Social Vulnerability Index	0.0206	0.0115 to 0.0293
Drive time to residency program (log)	−0.0097	−0.0191 to −0.0007
Drive time to any ophthalmologist (log)	0.0063	−0.0050 to 0.0172
Ophthalmologists per 100,000	−0.0031	−0.0126 to 0.0061
Optometrists per 100,000	−0.0135	−0.0225 to −0.0047
Population density (log)	−0.0028	−0.0203 to 0.0147
Rural-Urban Continuum Code	−0.0201	−0.0324 to −0.0072
Spatial dependence ρ	0.8265	0.7281 to 0.9112

Table 4: Social vulnerability coefficient under adjustment for external measures of diagnostic intensity, preventive-screening engagement, and insurance coverage. Each block is estimated on its own complete-case sample (13,035, 13,008 and 13,024 cells respectively), so each block carries its own base row; all three round to the same value.

Domain	Specification	β_{SVI}	Cluster SE	p
Diagnostic intensity	Base (block sample)	0.0353	0.0062	< 0.001
	+ Medicare imaging, E&M, tests per 1,000	0.0354	0.0063	< 0.001
Preventive screening	Base (block sample)	0.0353	0.0062	< 0.001
	+ Mammography screening	0.0206	0.0069	0.003
	+ Mammography and influenza vaccination	0.0217	0.0077	0.005
Coverage	Base (block sample)	0.0353	0.0062	< 0.001
	+ Uninsured rate	0.0335	0.0076	< 0.001
	+ Uninsured rate and dual-eligible share	0.0343	0.0091	< 0.001
Standard errors cluster-robust, clustered on county.				

Table 5: Medicaid expansion contrast with a built-in placebo. Expansion altered coverage only for adults under 65, so an equal contrast in the 65 to 84 band indicates state-level confounding rather than a coverage effect.

Age band	Expansion status	Cells	Counties	β_{SVI}	Cluster SE
40–64 years	Non-expansion	2,736	1,127	0.0163	0.0125
40–64 years	Expansion	3,822	1,780	0.0461	0.0065
65–84 years	Non-expansion	2,436	1,127	0.0122	0.0118
65–84 years	Expansion	3,307	1,788	0.0421	0.0060
Triple interaction SVI $\times$ expansion $\times$ age band: $\beta = 0.0023$, cluster-robust $p = 0.47$. Standard errors cluster-robust, clustered on county.					

4 Discussion

Among United States adults who already have diabetic retinopathy, those living in more socially vulnerable counties are more likely to have disease that has reached a vision-threatening stage. The association is modest in magnitude, on the order of 0.4 percentage points of VTDR share per standard deviation of social vulnerability, but it is consistent across racial and ethnic groups and age bands, is not explained by demographic composition, and survives adjustment for substantial spatial dependence.

This finding is distinct from the well-established observation that DR is more prevalent in disadvantaged populations. That association is driven substantially by the social patterning of diabetes itself. Ours is conditioned on disease: it concerns how far retinopathy has advanced among those who have it. Because the progression from mild non-proliferative disease to vision-threatening disease is the interval in which screening and timely referral operate, a social gradient in that quantity is more readily interpreted as a gradient in the timeliness and continuity of ophthalmic care than as a gradient in diabetes risk.

That the gradient is not narrower among Medicare-eligible adults is, in our view, the most policy-relevant result, and external data reinforce it. Neither the county uninsured rate nor the dual-eligible share explained the gradient, and an apparent steepening of the gradient in Medicaid expansion states proved spurious: the identical contrast appeared among adults aged 65 to 84, whose coverage expansion could not have altered. Whatever produces this gradient, it is not principally a coverage mechanism.

Two external tests point instead toward preventive-care engagement. Adjusting for direct claims measures of diagnostic intensity, Medicare imaging, evaluation and management, and test volume per 1,000 beneficiaries, left the gradient completely unchanged, even though those measures are themselves associated with the outcome in the expected direction. Adjusting for county mammography screening completion, by contrast, attenuated it by roughly 42%. Mammography has no ophthalmic content; what it captures is whether a population follows through on recommended preventive screening at all. The implication is that the failure is not ophthalmology-specific but sits in the general machinery of preventive care, which is consistent with the reported associations between adverse social determinants and eye care utilization.[6, 14, 16, 18] We note that mammog-

391 raphy is a female-only screen used here as a county-level index, so it is a marker of screening culture
392 rather than an individual-level mediator, and this analysis should be read as decomposition rather
393 than formal mediation.

394 The gradient also sits within a coherent pattern. Social vulnerability predicted preventable
395 hospital stays, an ambulatory-care-sensitive measure drawn from a different data system and a
396 different organ system, at 0.279 standard deviations per standard deviation, against 0.100 for the
397 vision-threatening share. The diabetic retinopathy result is therefore modest relative to other
398 county care quality gradients, but it is not anomalous.

399 We did not find support for the hypothesis that proximity to academic ophthalmology infras-
400 tructure is associated with less advanced disease. If anything, counties nearer to residency programs
401 showed slightly *higher* vision-threatening shares, an association that was small but persisted under
402 spatial adjustment. We interpret this pattern as more likely reflecting diagnostic intensity than
403 clinical benefit: where subspecialty capacity is concentrated, vision-threatening disease is more
404 completely identified and coded. This interpretation is consistent with the caution offered by the
405 authors of the underlying surveillance estimates, who noted that claims diagnoses are influenced
406 by access to ophthalmologists and that their model may mistake such factors for true variation in
407 prevalence.[15] A full account of these tests is given in the Appendix.

408 Our findings both corroborate and refine the clinic-based evidence. Xie et al. reported a
409 dose-response between neighborhood deprivation and proliferative disease among referral-center
410 patients.[20] We reproduce a social gradient in disease severity at national population scale, which
411 is meaningful corroboration given the different sampling frame. We depart from that work in three
412 respects. Conditioning on the presence of diabetic retinopathy separates progression from occur-
413 rence, and we find the gradient in progression to be several times that in occurrence among people
414 with diabetes, so the two are not interchangeable. Using the diabetes population as the explicit
415 denominator addresses the confounding by neighborhood diabetes prevalence that the earlier study
416 identified as an open limitation. And whereas that study found greater distance from the oph-
417 thalmology clinic associated with proliferative disease, we find no population-level counterpart; we
418 take up that divergence, and the referral selection that most plausibly explains it, in a companion
419 analysis of academic ophthalmology proximity using the same data.

420 Our second contribution is methodological and applies to any study using VEHSS county esti-

421 mates. Because those estimates are constructed by post-stratifying national demographic-specific
422 rates onto local composition, a county’s racial and age structure moves its estimate mechanically.
423 We find that roughly three-quarters of the apparent county-level social gradient disappears once
424 demographic composition is held fixed. Analyses that regress these county estimates on composite
425 social indices, most of which incorporate racial and ethnic composition, will therefore substantially
426 overstate local social gradients. We recommend within-stratum estimation as a routine safeguard.

427 4.1 Limitations

428 First, coverage of the race-stratified cells is a function of demographic group size, so the Black and
429 Hispanic stratum estimates rest on a more urban subset of counties than the non-Hispanic White
430 estimates.

431 Second, this is an ecological analysis of county aggregates; associations cannot be interpreted
432 at the individual level, and the social vulnerability of a county is not the social circumstance of any
433 patient within it.

434 Third, and most importantly, the outcome is derived from modeled estimates built on diagnosed
435 disease in Medicare and Medicaid claims. Disease that is never diagnosed does not enter. If under-
436 detection is more common in disadvantaged counties, and if it disproportionately removes mild cases
437 from the denominator, the VTDR share would rise for reasons unrelated to true progression. We
438 addressed this with direct claims measures of county diagnostic intensity rather than provider counts
439 alone, and the gradient was unchanged by their inclusion, which is reassuring but not dispositive:
440 those measures index all-cause diagnostic activity rather than retinal screening specifically. Residual
441 detection bias remains the principal threat, and its likely direction is to overstate our finding.

442 Fourth, VEHSS county random effects were themselves estimated controlling for ophthalmolo-
443 gists per capita. Including provider density as a covariate is therefore partly a second adjustment
444 for a factor already partially removed from the outcome, which will tend to attenuate the estimated
445 role of supply.

446 Fifth, drive times model private vehicle travel on the road network. They ignore traffic, are
447 unreliable across ferry routes, and do not represent public transportation at all. Fourteen island
448 and remote counties in Alaska and Hawaii had no routable path and are excluded from access
449 analyses.

Sixth, indicator vintages are not perfectly aligned. DR estimates are for 2021, SVI is constructed from 2016–2020 data, and workforce counts are 2021. The glaucoma comparison reported in the Appendix is published only for 2022.

Seventh, Puerto Rico is absent from the VEHSS source and is therefore unrepresented here, as are any counties failing the completeness criteria.

4.2 Conclusion

Social vulnerability is associated with more advanced diabetic retinopathy among people who already have the disease, independent of demographic composition, provider supply, and spatial structure, and undiminished by Medicare eligibility. Efforts to reduce preventable vision loss may be better directed at the timeliness and continuity of retinal screening within disadvantaged communities than at the geographic placement of academic ophthalmology infrastructure.

References

- [1] Ahmed A, Ali M, Dun C, et al. Geographic distribution of US ophthalmic surgical subspecialists. *JAMA Ophthalmol.* 2025;143(2):117–124.
- [2] American Academy of Ophthalmology. *Diabetic Retinopathy Preferred Practice Pattern.* 2024.
- [3] Berkowitz ST, Finn AP, Parikh R, Kuriyan AE, Patel S. Ophthalmology workforce projections in the United States, 2020 to 2035. *Ophthalmology.* 2024;131(2):133–139.
- [4] Centers for Disease Control and Prevention. Vision and Eye Health Surveillance System: composite prevalence estimates. Available at <https://data.cdc.gov/resource/qeru-k2y2>. Accessed August 2026.
- [5] Centers for Disease Control and Prevention / Agency for Toxic Substances and Disease Registry. Social Vulnerability Index 2020, county level. Available at <https://svi.cdc.gov>. Accessed August 2026.
- [6] Chaudhury AS, Ige M, Marwah S, et al. Race, social determinants of health, and the quality of diabetic eye care. *JAMA Ophthalmol.* 2024;142(10):961–970.
- [7] Feng PW, Ahluwalia A, Feng H, Adelman RA. National trends in the United States eye care workforce from 1995 to 2017. *Am J Ophthalmol.* 2020;218:128–135.
- [8] University of Wisconsin Population Health Institute. County Health Rankings & Roadmaps 2021 analytic data. Available at <https://www.countyhealthrankings.org>. Accessed August 2026.
- [9] Centers for Medicare & Medicaid Services. Medicare Geographic Variation by National, State & County, 2021. Available at <https://data.cms.gov>. Accessed August 2026.
- [10] Kaiser Family Foundation. Status of State Medicaid Expansion Decisions. Available at <https://www.kff.org/medicaid/>. Accessed August 2026.
- [11] Health Resources and Services Administration. Area Health Resources Files, 2022–2023 release. Available at <https://data.hrsa.gov>. Accessed August 2026.

- [12] Jani PD, Forbes L, Choudhury A, et al. Evaluation of diabetic retinal screening and factors for ophthalmology referral in a telemedicine network. *JAMA Ophthalmol.* 2017;135(7):706–714.
- [13] Lee D. CARBayes: an R package for Bayesian spatial modeling with conditional autoregressive priors. *J Stat Softw.* 2013;55(13):1–24.
- [14] Liu Y, Zupan NJ, Shiyanbola OO, et al. Factors influencing patient adherence with diabetic eye screening in rural communities: a qualitative study. *PLoS One.* 2018;13(11):e0206742.
- [15] Lundeen EA, Burke-Conte Z, Rein DB, et al. Prevalence of diabetic retinopathy in the US in 2021. *JAMA Ophthalmol.* 2023;141(8):747–754.
- [16] Ravindranath R, Bernstein IA, Fernandez KS, Ludwig CA, Wang SY. Social determinants of health and perceived barriers to care in diabetic retinopathy screening. *JAMA Ophthalmol.* 2023;141(12):1161–1171.
- [17] Soares RR, Mokhashi N, Sharpe J, et al. Patient accessibility to eye care in the United States. *Ophthalmology.* 2023;130(4):354–360.
- [18] Taccheri C, Jordan J, Tran D, et al. The impact of social determinants of health on eye care utilization in a national sample of people with diabetes. *Ophthalmology.* 2023;130(10):1037–1045.
- [19] US Census Bureau. Centers of Population by County, 2020 Census. Available at <https://www2.census.gov/geo/docs/reference/cenpop2020/>. Accessed August 2026.
- [20] Xie WY, Rustam Z, Tran D, et al. Association of neighborhood socioeconomic disadvantage with proliferative diabetic retinopathy. *Ophthalmol Retina.* 2025;9(2):e1–e9. doi:10.1016/j.oret.2024.10.012
- [21] Weng CY, Maguire MG, Flaxel CJ, et al. Effectiveness of conventional digital fundus photography-based teleretinal screening for diabetic retinopathy and diabetic macular edema: a report by the American Academy of Ophthalmology. *Ophthalmology.* 2024;131(8):927–942.